

6 Lipids, proteins and polysaccharides are biological molecules that can be described as macromolecules.

(a) Explain why lipids, proteins and polysaccharides can be described as macromolecules, but only proteins and polysaccharides can be described as polymers.

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..... [2]

(b) Table 6.1 lists five biological molecules and shows three types of bond.

Table 6.1

biological molecule	peptide bond	glycosidic bond	ester bond
amylopectin			
cellulose			
collagen			
glycogen			
triglyceride			

(i) Complete Table 6.1 to show the type of bond present in each biological molecule listed.

Use **one** tick (✓) in each row to indicate the presence of a bond. [3]

(ii) With reference to Table 6.1, state the **two** biological molecules that are **most similar** to each other **and** explain your answer.

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..... [2]

- (c) Mucin is an example of a glycoprotein. Mucin is present in the mucus that lines the surface of the ciliated epithelium of the gas exchange system.

Name **two** structures in the gas exchange system that produce mucus.

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..... [2]

- (d) A large proportion of substances that are known as antigens are proteins, glycoproteins and polysaccharides.

Outline **other** features that would help to identify a substance as an antigen.

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..... [3]

[Total: 12]